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Roll No.

**B.TECH-Third Semester(Backlog)-Course work
END-SEMESTER THEORY EXAMINATION, NOV-DEC,2024**

Course Code: CMMTC04

Course Title: *ENGINEERING MATHEMATICS*

Time: 3:00 Hours

Max. Marks: 50

Note: Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO														
Q1	Attempt any 2 parts of the following.																
1a	(1) Find the Laplace transform of $t^2 e^{-2t} \cos t$. (2) Find the inverse Laplace transform of $\frac{1}{(t^2 + 9)^2}$.	5	CO1														
1b	Express the function $f(t) = \begin{cases} 2t, & \text{if } 0 < t \leq 5 \\ 10, & \text{if } 5 < t \end{cases}$ in terms of unit step function and then find its Laplace transform.	5	CO1														
1c	Solve $y'' + y = x \cos 2x$, $y(0) = 0$, $y'(0) = 0$, using Laplace transform.	5	CO1														
Q2	Attempt any 2 parts of the following.																
2a	Find the Fourier series of the function $f(x) = \begin{cases} x, & \text{if } 0 < x \leq \pi \\ 2\pi - x, & \text{if } \pi < x < 2\pi \end{cases}$ given $f(x) = f(x + 2\pi)$.	5	CO2														
2b	Find the Fourier cosine series for $f(x) = x \sin x$, $0 < x < \pi$.	5	CO2														
2c	Find the Fourier series upto second harmonic from the following data:	5	CO2														
	<table><tr><td>x</td><td>0°</td><td>60°</td><td>120°</td><td>180°</td><td>240°</td><td>300°</td></tr><tr><td>y</td><td>1.98</td><td>1.30</td><td>1.05</td><td>1.30</td><td>-0.88</td><td>-0.25</td></tr></table>	x	0°	60°	120°	180°	240°	300°	y	1.98	1.30	1.05	1.30	-0.88	-0.25		
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y	1.98	1.30	1.05	1.30	-0.88	-0.25											

Q. No.	Question	Marks	CO
Q3	Attempt any 2 parts of the following.		
3a	Solve the one-dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(x, 0) = 3 \sin n\pi x$, $u(0, t) = u(l, t) = 0$, with $0 < x < l$.	5	CO3
3b	Let $u(x, t)$ be the solution of the initial boundary value problem $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, $-\infty < x < \infty$, $t > 0$, $u(x, 0) = x^4$, $u_t(x, 0) = \sin x$. Then find the value of $u(2, 2)$.	5	CO3
3c	Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to boundary conditions $u(0, y) = \sin y$, $u(x, y) \rightarrow 0$ as $x \rightarrow \infty$.	5	CO3
Q4	Attempt any 2 parts of the following.		
4a	Show that the function defined by $f(z) = \sqrt{ xy }$ satisfies Cauchy-Riemann equation at the origin but is not analytic at that point.	5	CO4
4b	Evaluate $\int_C \frac{z-3}{z^2-2z}$ where C is the circle $ z+1-i = 2$.	5	CO4
4c	Using Cauchy-residue theorem evaluate $\int_C \frac{3z^3+2}{(z-1)(z^2+9)}$ where C is the circle $ z = 4$.	5	CO4
Q5	Attempt any 2 parts of the following.		
5a	Using Gauss Divergence theorem, evaluate $\int \int \vec{F} \cdot \hat{n} \cdot ds$ where $\vec{F} = (2x+3z)\hat{i} - (xz+y)\hat{j} + (y^2+2z)\hat{k}$ and S is the surface of the sphere $x^2+y^2+z^2 = 9$.	5	CO5
5b	Show that the vector field defined by the vector function $\vec{F} = xyz(yz\hat{i} + xz\hat{j} + xy\hat{k})$ is conservative and find the scalar potential function ϕ .	5	CO5
5c	Verify Green's theorem for the vector field $\vec{F} = (x-y)\hat{i} + x\hat{j}$ and the region R bounded by the unit circle $r(t) = \cos(t)\hat{i} + \sin(t)\hat{j}$, $0 \leq t \leq 2\pi$.	5	CO5